

A Novel Base-Catalyzed Stereoselective Synthesis of Trifluoromethylated Dienedinitriles

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Abstract: The trifluoromethylated γ -cyano- β,γ -unsaturated nitrile **4** was prepared by the reaction of bromo(cyanomethyl)zinc with trifluoroacetylated phosphonate **3** in 67% yield ($E:Z = 60:40$). The E - and Z -isomers obtained were easily separated by column chromatography. Treatment of a catalytic amount of t -BuOK (20 mol%) with a mixture of (Z)-**4** and N -toluenesulfonylimines **5** gave trifluoromethylated dienedinitriles **6** in 68–83% yields.

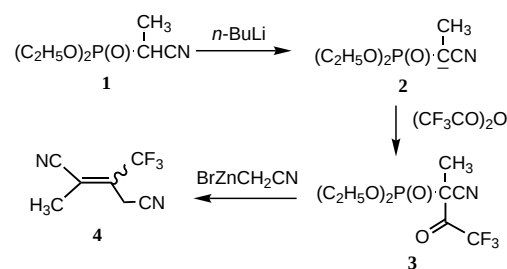
Key words: trifluoromethylated dienedinitriles, γ -cyano- β,γ -unsaturated nitrile, trifluoroacetylated phosphonate, organozinc reagent, stereoselective synthesis

1,5-Dinitriles are 1,5-bifunctional compounds, which are a useful class of substrates or precursors in aldol-type condensation and other annulation reactions.¹ The cyano group is an important functional group in organic transformations and the cyano-containing compounds are useful intermediates for the synthesis of heterocyclic compounds.² The trifluoromethylated heterocyclic compounds have been applied increasingly in the field of pharmaceuticals and agrochemicals,³ since the introduction of a trifluoromethyl group into organic compounds often dramatically changes their structure, stability, reactivity and biological activity of the resulting compounds.⁴ However, to the best of our knowledge a method for the synthesis of trifluoromethylated dienedinitriles has not been reported in the literature. Therefore, development of an efficient method for the synthesis of the title compounds would be valuable.

In our previous papers, the double olefination methodology was used to synthesize 3-trifluoromethyl (2*Z*,4*E*)-pentadienenitriles,⁵ and a stereocontrolled method was used to synthesize (2*E*,4*E*)- or (2*Z*,4*E*)-pentadienenitriles.⁶ As part of our continuing investigation to explore new synthetic methodologies for carbon–carbon double bond formation,⁷ we report herein a novel base-catalyzed stereoselective methodology for the synthesis of trifluoromethylated (2*Z*,4*Z*)-4-cyanopentadienenitriles exclusively or predominately.

The phosphoryl-stabilized carbanion **2**, generated from diethyl (1-cyanoethyl)phosphonate (**1**) and butyllithium in THF, was acylated by the addition of trifluoroacetic anhydride to give the trifluoroacetylated phosphonate **3**. The

phosphonate **3** was reacted in situ with bromo(cyanomethyl)zinc with elimination of phosphonic acid anion to afford the trifluoromethylated γ -cyano- β,γ -unsaturated nitrile **4** in 67% yield ($E:Z = 60:40$) (Scheme 1). The E - and Z -isomers obtained were easily separated by column chromatography.



Scheme 1

¹⁹F NMR data reported for 4,4,4-trifluoro-3-phenylbut-2-enenitrile in the literature⁸ assert that the trifluoromethyl group *cis* with respect to the cyano group (Z -isomer) appears at -15.6 ppm, while that *trans* with respect to the cyano group (E -isomer) at -10.9 ppm (Figure 1). Based on these results, the CF_3 group in compound **4** with the chemical shift at -13.4 ppm has been assigned as the E -isomer, and the one at -17.4 ppm as the Z -isomer.

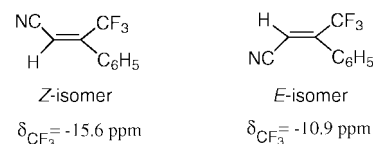
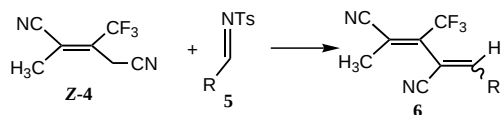


Figure 1 Reported chemical shifts for CF_3 in (E)- and (Z)-4,4,4-trifluoro-3-phenylbut-2-enenitrile

Compound (Z)-**4** was used as the starting material for the synthesis of the title compounds. Treatment of a catalytic amount of t -BuOK (20 mol%) with a mixture of (Z)-**4** and N -toluenesulfonylimines **5** gave trifluoromethylated dienedinitriles **6** in 68–83% yields (Scheme 2). The results are summarized in Table 1.

For the assignment of the configuration of the products **6**, we performed an X-ray crystallographic analysis of compound **6e** (major portion). The crystal structure shows that the cyano group is *cis* with respect to the trifluoromethyl group ($2Z$) and the other cyano group across the other



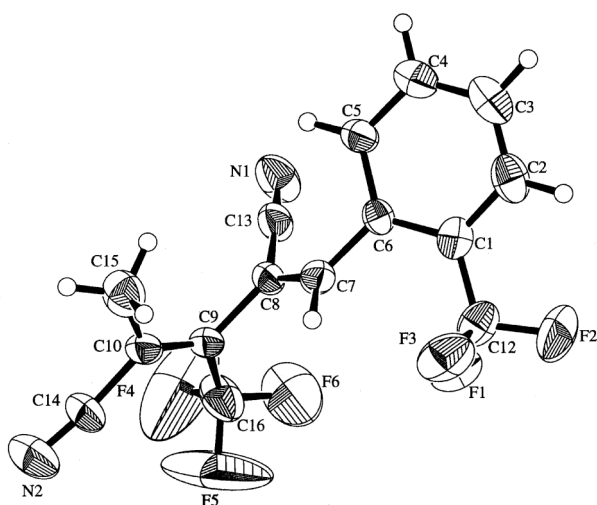
Scheme 2

Table 1 Preparation of Trifluoromethylated Dienedinitriles

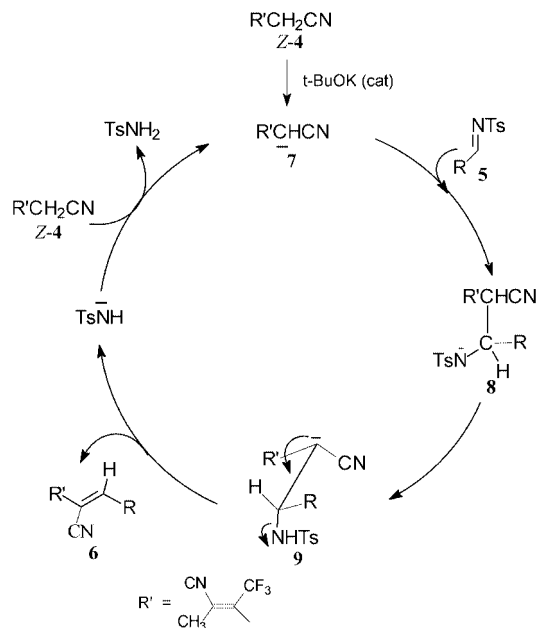
Product	R	Yield (%) ^a	2Z,4Z:2Z,4E ^b
6a	4-FC ₆ H ₄	81	100:0
6b	2,4-Cl ₂ C ₆ H ₃	75	100:0
6c	4-MeC ₆ H ₄	70	100:0
6d	4-NO ₂ C ₆ H ₄	78	100:0
6e	2-CF ₃ C ₆ H ₄	68	94:6
6f	4-ClC ₆ H ₄	79	92:8
6g	4-MeOC ₆ H ₄	83	82:18

^a Isolated yield.^b E/Z-Ratio of the isolated product.

double bond is also *cis* with respect to the 2-trifluoromethylphenyl group (4Z) (Figure 2). Hence, the configuration of products could be ascertained on the basis of the crystal structure.

**Figure 2** Crystal structure for (2Z,4Z)-4-cyano-3-trifluoromethyl-5-(2-trifluoromethylphenyl)-2-methylpentadienenitrile [(2Z,4Z)-**6e**]

The catalytic reaction mechanism may be rationalized as given in Scheme 3. The carbanion **7**, generated from (Z)-**4** and *t*-BuOK reacts with *N*-toluenesulfonylimine **5** to give the adduct **8**. After hydrogen transfer, **8** affords **9**. Elimination of *p*-toluenesulfonamide anion from **9** furnishes the product **6**. The *p*-toluenesulfonamide anion abstracts a proton from (Z)-**4** to give *p*-toluenesulfonamide and regenerates the carbanion **7**. Thus, the catalytic cycle is completed.



Scheme 3

The stereochemical results may be rationalized as shown in Scheme 4. The reaction is initiated by the nucleophilic attack of the carbanion **7** on the imine carbon forming **8a** and **8b**, which after hydrogen transfer affords **9a** and **9b**, respectively. Space filling models indicate that⁹ the relative size of groups is R (aryl) > R' > CN > H. Since the intermediate **9a** involves an orientation of two pairs of 'small'/'large' substituents (R'/H, R/CN), this conformer should be favored relative to the stereoisomer **9b** which contains unfavorable 'large'/'large' (R/R') non-bonded interactions. Therefore, in our case, the stereoselectivity is a function of the conformational equilibrium of the adduct. Each of these intermediates affords (2Z,4Z)-**6** or (2Z,4E)-**6**, respectively, via an *anti*-elimination. In our case, formation of **9a** will be favored over **9b** and the 2Z,4Z-isomer was obtained exclusively or predominantly.

In conclusion, a novel base-catalyzed stereoselective route to trifluoromethylated (2Z,4Z)-4-cyanopentadienenitriles has been designed. It is noteworthy that the use of *N*-toluenesulfonylimines as substrates makes the reaction catalytic to prepare the conjugated alkene moiety. Thus, this methodology provides a simple and convenient synthesis of the title compounds from easily accessible starting materials. The title compounds are not easily available with existing synthetic methods.

Petroleum ether used had bp 60–90 °C. Bps and Mps are uncorrected. The IR spectra of liquid products were obtained as films and those of solid products as KBr disks on a Digilab FTS-20E spectrometer. ¹H NMR spectra were recorded in CDCl₃ on a Bruker AM-300 (300 MHz) spectrometer (δ values in ppm from tetramethylsilane, *J*-values are given in Hz). ¹⁹F NMR spectra were taken on a Varian EM-360 (60 MHz) spectrometer. [δ values from external trifluoroacetic acid (TFA), in CDCl₃, positive for upfield shifts]. Mass spectra were measured on a Finnigan GC-MS-4021 mass

(2Z,4Z)-4-Cyano-3-trifluoromethyl-2-methyl-5-(4-nitrophenyl)pentadienenitrile [(2Z,4Z)-6d]

Yield: 78%; oil.

IR (KBr): 2220, 1600, 1530, 1350, 1330, 1220, 1170, 1140 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 8.36 (dd, *J* = 7.0, 1.6 Hz, 2 H), 8.02 (dd, *J* = 8.7, 1.7 Hz, 2 H), 7.46 (s, 1 H), 2.37 (q, *J* = 2.2 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -17.0 (s, 3 F).MS: *m/z* (%) = 307 (M⁺, 18), 292 (54), 261 (12), 238 (74), 221 (22), 192 (69), 149 (100).HRMS: *m/z* Calcd for C₁₄H₈F₃N₃O₂ (307.23): 307.0569. Found: 307.0519.**(2Z,4Z)-4-Cyano-3-trifluoromethyl-5-(2-trifluoromethylphenyl)-2-methylpentadienenitrile [(2Z,4Z)-6e]**

Yield: 64%; mp 120–122 °C.

IR (KBr): 2220, 1640, 1440, 1380, 1320, 1210, 1170, 1170, 1130 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 7.98 (d, *J* = 7.6 Hz, 1 H), 7.82 (d, *J* = 7.6 Hz, 1 H), 7.74–7.62 (m, 2 H), 7.59 (s, 1 H), 2.28 (q, *J* = 1.9 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -18.6 (s, 3 F), -16.0 (s, 3 F).MS: *m/z* (%) = 331 (M⁺ + 1, 18), 315 (64), 295 (44), 261 (100), 234 (28), 214 (10).Anal. Calcd for C₁₅H₈F₆N₂ (330.23): C, 54.56; H, 2.44; N, 8.48. Found: C, 54.82; H, 2.31 N, 8.53.**(2Z,4E)-4-Cyano-3-trifluoromethyl-5-(2-trifluoromethylphenyl)-2-methylpentadienenitrile [(2Z,4E)-6e]**

Yield: 4%; oil.

IR (KBr): 2230, 1600, 1580, 1490, 1320, 1230, 1130, 1040 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 8.02 (d, *J* = 7.4 Hz, 1 H), 7.80 (d, *J* = 7.6 Hz, 1 H), 7.73 (s, 1 H), 7.72–7.62 (m, 2 H), 2.36 (q, *J* = 2.3 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -18.6 (s, 3 F), -18.0 (s, 3 F).MS: *m/z* (%) = 330 (M⁺, 5), 315 (73), 311 (8), 295 (50), 261 (100), 241 (21), 234 (28).Anal. Calcd for C₁₅H₈F₆N₂ (330.23): C, 54.56; H, 2.44; N, 8.48. Found: C, 54.27; H, 2.17 N, 8.39.**(2Z,4Z)-5-(4-Chlorophenyl)-4-cyano-3-trifluoromethyl-2-methylpentadienenitrile [(2Z,4Z)-6f]**

Yield: 72%; mp 96–98 °C.

IR (KBr): 2200, 1630, 1600, 1590, 1490, 1380, 1330, 1210, 1170, 1140, 1100 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 7.83 (d, *J* = 6.8 Hz, 2 H), 7.49 (d, *J* = 6.6 Hz, 2 H), 7.12 (s, 1 H), 2.27 (q, *J* = 2.0 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -16.0 (s, 3 F).MS: *m/z* (%) = 298 (M⁺ + 2, 32), 296 (M⁺, 81), 281 (100), 261 (98), 241 (25), 227 (90), 192 (59).Anal. Calcd for C₁₄H₈ClF₃N₂ (296.67): C, 56.68; H, 2.72; N, 9.44. Found: C, 56.41; H, 2.44; N, 9.46.**(2Z,4E)-5-(4-Chlorophenyl)-4-cyano-3-trifluoromethyl-2-methylpentadienenitrile [(2Z,4E)-6f]**

Yield: 7%; oil.

IR (KBr): 2220, 1630, 1600, 1590, 1490, 1330, 1210, 1170, 1140, 1100 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 7.83 (d, *J* = 8.6 Hz, 2 H), 7.55 (d, *J* = 8.8 Hz, 2 H), 7.29 (s, 1 H), 2.34 (q, *J* = 2.5 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -19.8 (s, 3 F).MS: *m/z* (%) = 298 (M⁺ + 2, 26), 296 (M⁺, 73), 281 (99), 261 (100), 227 (92), 200 (26), 192 (59).Anal. Calcd for C₁₄H₈ClF₃N₂ (296.67): C, 56.68; H, 2.72; N, 9.44. Found: C, 56.61; H, 2.46 N, 9.51.**(2Z,4Z)-4-Cyano-3-trifluoromethyl-5-(4-methoxyphenyl)-2-methylpentadienenitrile [(2Z,4Z)-6g]**

Yield: 68%; oil.

IR (neat): 2220, 1600, 1570, 1510, 1430, 1320, 1270, 1210, 1180, 1140, 1030 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 7.89 (dd, *J* = 7.0, 2.0 Hz, 2 H), 7.06 (s, 1 H), 7.00 (dd, *J* = 6.9, 2.0 Hz, 2 H), 3.90 (s, 3 H), 2.26 (q, *J* = 1.9 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -16.3 (s, 3 F).MS: *m/z* (%) = 292 (M⁺, 100), 277 (68), 257 (57), 223 (26), 208 (21), 196 (10).Anal. Calcd for C₁₅H₁₁F₃N₂O (292.26): C, 61.65; H, 3.79; N, 9.59. Found: C, 61.57; H, 3.81; N, 9.59.**(2Z,4E)-4-Cyano-3-trifluoromethyl-5-(4-methoxyphenyl)-2-methylpentadienenitrile [(2Z,4E)-6g]**

Yield: 15%; oil.

IR (neat): 2220, 1600, 1510, 1440, 1310, 1330, 1270, 1180, 1140, 1030 cm⁻¹.¹H NMR (CDCl₃/TMS): δ = 7.90 (d, *J* = 8.4 Hz, 2 H), 7.26 (s, 1 H), 6.96 (d, *J* = 8.4 Hz, 2 H), 3.88 (s, 3 H), 2.31 (q, *J* = 2.3 Hz, 3 H).¹⁹F NMR (CDCl₃/TFA): δ = -19.4 (s, 3 F).MS: *m/z* (%) = 293 (M⁺ + 1, 42), 292 (M⁺, 100), 277 (60), 223 (26), 208 (19).Anal. Calcd for C₁₅H₁₁F₃N₂O (292.26): C, 61.65; H, 3.79; N, 9.59. Found: C, 61.46; H, 3.66 N, 9.47.**Crystal Structure Determination of 6e¹⁰**

Crystal Data: C₁₅H₈F₆N₂, *M* = 330.23, orthorhombic, space group P2₁2₁1 (no. 19), *a* = 8.928(2), *b* = 21.325(2), *c* = 7.766(2) Å, *V* = 1478.6(5) Å³, *Z* = 4, *D*_{calc} = 1.483 g cm⁻³, λ(MoKα) = 0.71069 Å, μ = 1.42 cm⁻¹, *T* = 293.0 K, prismatic crystal, 0.20 × 0.20 × 0.30 mm.

Data Correction and Processing: Data were measured on a Rigaku AFC7R diffractometer with graphite-monochromated MoKα radiation using the ω–2θ scan technique to a maximum 2θ-value of 50.0°. 1992 reflections were collected. The data were corrected for Lorentz and polarization factors. A correction for secondary extinction was also applied.

Structure Solution and Refinement: The structure was resolved by direct method and expanded using Fourier techniques. The non-hydrogen atoms were refined anisotropically. Hydrogen atoms were included but not refined. The final cycle of fullmatrix least-squares refinement was based on 1394 observed reflections and 209 variable parameters. The final *R*- and *R*_w-values are 0.063 and 0.055, respectively. All calculations were performed using the TEXAN crystallographic software package from Molecular Structure Corporation.

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- (10) Crystallographic data of **6e** have been deposited with the Cambridge Crystallographic Data Centre as supplementary publication No. CCDC-208198(**6e**). Copies of the data can be obtained, free of charge, on application to CCDC, 12 Union Road, Cambridge CB2 1EZ, UK (fax: +44 1223 336033 or e-mail: deposit@ccdc.cam.ac.uk).